

I have updated the full PRD to incorporate both new fraud-prevention requirements and the dedicated Admin Settings Menu structure.

Here is the revised spec for your MD and CTO:

Product Requirements Document (PRD): Admin PIN & Terminal Authentication System

1. Executive Summary

This document outlines the product and technical specifications for introducing an **Agent Setup Verification Code**, an **Admin PIN** authentication system, and an extended **Admin Settings Menu** to our terminal management system. The architecture is explicitly designed to mitigate fraud during terminal delivery by ensuring that terminals linked to live corporate accounts cannot be initialized without a one-time agent setup code.

2. Security & Fraud Mitigation Objectives

- **Delivery & Unboxing Fraud Prevention:** Block unauthorized activation of delivered terminals by requiring a dynamic One-Time Setup Code delivered via SMS to the authorized terminal owner/agent.
- **Administrative Access Control:** Restrict sensitive device configurations and transaction credentials behind a mandatory Admin PIN.
- **Centralized Admin Management:** Group administrative actions (Admin PIN resets and Transaction PIN changes) into a dedicated Admin Settings page.
- **Multi-Channel Recovery Controls:** Retain robust fallback mechanisms (SMS, WhatsApp, and Tier-2 Neptune Support bypass) for forgotten Admin PINs.

3. End-to-End User Workflows & System Behavior

3.1. First-Time Terminal Initialization (Agent Setup Verification)

To prevent delivery interceptors or unauthorized handlers from configuring terminals pre-delivery, the terminal is locked upon first boot.

- **Pre-requisite:** Upon successful backend terminal registration and assignment to a customer's live corporate account, the system auto-generates a dynamic, one-time **Setup Code / OTP** and transmits it via SMS to the registered agent/owner's phone.
- **Initial Screen (Locked State):** Upon first Neptune MFB application boot, the terminal exclusively renders the **Enter Setup Code** screen.
 - This screen persists across reboots until valid code entry.
 - No administrative features, transaction screens, or PIN creation dialogs are accessible prior to this step.
- **Validation & Next Step:** Once the agent inputs the correct Setup Code, the terminal completes device registration and immediately triggers the initial **Admin PIN Creation Flow** (Section 3.2).
- **Subsequent Boots:** Subsequent application boots completely bypass the Setup Code screen.

3.2. First-Time Admin PIN Setup

Immediately following successful Setup Code verification, the terminal prompts the agent to establish their master Admin PIN.

- **Trigger:** Completion of Step 3.1.
- **UI Components:** Modal/dialog containing **Enter Admin PIN** and **Confirm Admin PIN** fields.
- **Backend Integration:** The backend securely hashes and associates the Admin PIN with the Terminal ID.

3.3. Daily Access to Admin Settings

- **Trigger:** User taps the **Settings** icon on the main dashboard.

- **UI Components:** Modal pop-up requesting Admin PIN containing:
 - PIN Entry Input
 - **Confirm** Button
 - **Cancel** Button
 - **Forgot Password?** Link
 - Helper Text: *"Contact support if you do not receive an OTP."*
- **Navigation:** Upon successful Admin PIN authentication, the terminal navigates the user to the dedicated **Admin Settings Menu** (Section 3.4).

3.4. Admin Settings Menu Structure

Once authenticated via the Admin PIN, the user lands on the primary **Admin Settings** dashboard. This screen hosts two core management functions:

1. **Reset Admin PIN:** Allows the user to execute a complete update/reset of their master Admin PIN.
2. **Change Transaction PIN:** Navigates the user to update the terminal's operational Transaction PIN used for processing card/payment transactions.

3.5. Admin PIN Recovery Workflow (SMS, WhatsApp, & Support Bypass)

If an owner forgets their Admin PIN, they initiate self-service or support-assisted recovery:

1. **Automated OTP Recovery (SMS/WhatsApp):**
 - User taps **Forgot Password?** on the Admin PIN prompt.
 - An OTP is dispatched via SMS. Upon entry, the user is navigated directly to the **Reset Admin PIN** screen.
2. **Support-Assisted Recovery & WhatsApp Verification Fallback:**

- If SMS OTP fails, the user clicks **Contact Support**. The support phone number is displayed on screen.
- **Support Protocol:** Support verifies if the registered phone number is active on WhatsApp:
 - **If Active on WhatsApp:** Support dispatches the OTP via WhatsApp for standard entry.
 - **If Inactive / Non-WhatsApp Number:** Support initiates a remote **Support Bypass** on the backend management portal.
- **Audit Requirement:** Support agents must input their **Support Agent Name / ID** prior to granting bypass authorization.
- **Terminal Behavior Post-Bypass:** User taps Admin Settings and proceeds with an empty PIN field (or taps **Proceed**) to enter the **Reset Admin PIN** flow.
- **Bypass Expiration Rules:** Active bypass flags automatically expire **6 minutes** post-activation (configurable) or instantly upon successful PIN update.

4. Technical & Security Specifications

Parameter	Technical Specification
Terminal Setup Code (OTP)	One-time activation code sent via SMS upon customer terminal assignment. Persists as sole screen until verified. Single-use only.
Default / Master Fallback PIN	The current default/legacy Admin PIN is reassigned strictly as the internal Neptune Support PIN for internal field maintenance.

Bypass Expiration Window	Default TTL: 6 Minutes (configurable value). Automated backend TTL index revokes flag post-expiration.
Audit Trail Requirements	Backend logs must record: Terminal ID , Action Type (Setup Code Generation, Bypass Activation) , Support Agent Identifier , Timestamps , and Status .

5. Screen Hierarchy & Navigation Flow

- **Screen 1: Agent Setup Code Entry** (First boot only — blocks all dashboard access)
- **Screen 2: Initial Admin PIN Creation** (Immediately follows Screen 1)
- **Screen 3: Main Terminal Dashboard** (Contains Admin Settings Icon)
- **Screen 4: Admin PIN Authentication Modal** (Prompts PIN + Forgot Password link)
- **Screen 5: Admin Settings Page** (Hub containing: **Reset Admin PIN** & **Change Transaction PIN**)
- **Screen 6: OTP Verification & Support Display** (Recovery flows)
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